Generic Degree Three in $JC(3)$: Collision Geometry and Galois Group S_3

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Abstract

Alpöge's explicit counterexample to the complex three-dimensional Jacobian conjecture has generic degree three [Gao26, Theorem 3.3]. Motivated by this example, we define an algebraic obstruction to invertibility and study it in generic degree three. For a polynomial map $F\colon \mathbb{A}^n_{\mathbb{C}} \to \mathbb{A}^n_{\mathbb{C}}$, let R_F be the scheme of pairs (x, y) satisfying $F(x) = F(y)$. In $S = \mathbb{C}[x_1, \dots, x_n, y_1, \dots, y_n]$, the collision scheme and its diagonal are defined respectively by

$$I_R(F) = \big(F_i(x) - F_i(y)\big)_{i=1}^n, \qquad I_\Delta = (x_i - y_i)_{i=1}^n.$$

Since $I_R(F) \subseteq I_\Delta$, the quotient

$$\mathrm{Obs}(F) := I_\Delta / I_R(F) \subseteq C_F := S/I_R(F)$$

defines the diagonal inside R_F. Over $\mathbb{C}$, the map F is a polynomial automorphism precisely when $I_R(F) = I_\Delta$, equivalently when $\mathrm{Obs}(F) = 0$. The Jacobian conjecture $JC(n)$ asserts this equality for every map with $\det JF \in \mathbb{C}^\times$. The coordinate functions of F generate

$$B = \mathbb{C}[F_1, \dots, F_n] \subseteq A = \mathbb{C}[x_1, \dots, x_n],$$

and hence a function-field extension $K = \mathrm{Frac}(B) \subseteq L = \mathrm{Frac}(A)$. For $[L : K] = 3$, F has no polynomial inverse. In dimension three, every map of this degree satisfying the Jacobian condition is a counterexample to $JC(3)$.

For every such counterexample, the generic collision algebra decomposes as

$$K \otimes_B C_F \simeq_K L \times N,$$

where N is the normal closure of L/K. Consequently, the normal closure of the function-field extension attached to every generic-degree-three counterexample to $JC(3)$ has Galois group S_3.

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1 Introduction

The complex Jacobian conjecture $JC(n)$ asks whether every polynomial map

$$F = (F_1, \dots, F_n) \colon \mathbb{A}^n_{\mathbb{C}} \longrightarrow \mathbb{A}^n_{\mathbb{C}}, \qquad \det JF \in \mathbb{C}^\times,$$

has a polynomial inverse. Such an F is called a Keller map. The Jacobian condition makes F an étale morphism; the conjecture asks whether this local condition forces global invertibility.

Alpöge's explicit three-dimensional counterexample [Alp26] is

$$F_0(x, y, z) = \begin{pmatrix} (1+xy)^3 z + y^2(1+xy)(4+3xy) \\ y + 3x(1+xy)^2 z + 3xy^2(4+3xy) \\ 2x - 3x^2 y - x^3 z \end{pmatrix}.$$

It satisfies $\det JF_0 = -2$ and has generic degree three [Gao26, Theorem 3.3]. The determinant identity and failure of injectivity have been independently formalized in Isabelle/HOL [RHdQ26]. This example motivates a uniform structural question: what is the generic collision algebra of a three-dimensional Keller map of generic degree three?

Put

$$S = \mathbb{C}[x_1, \dots, x_n, y_1, \dots, y_n].$$

The equations

$$F_i(x) - F_i(y) = 0 \qquad (1 \le i \le n)$$

generate the collision ideal, while the equations of the diagonal generate the diagonal ideal:

$$I_R(F) = \big(F_i(x) - F_i(y)\big)_{i=1}^n, \qquad I_\Delta = (x_i - y_i)_{i=1}^n.$$

Thus the collision scheme is

$$R_F = \operatorname{Spec} C_F, \qquad C_F = S/I_R(F).$$

Its complex points are precisely the pairs (x, y) with $F(x) = F(y)$; equivalently, R_F is the fiber square of F. It contains the diagonal Δ, consisting of the pairs (x, x). Since every collision equation vanishes on the diagonal,

$$I_R(F) \subseteq I_\Delta.$$

Its image in C_F is the ideal

$$\mathrm{Obs}(F) := I_\Delta / I_R(F) \subseteq C_F,$$

which defines the diagonal inside R_F. By Ax–Grothendieck,

$$I_R(F) = I_\Delta \iff \mathrm{Obs}(F) = 0 \iff R_F = \Delta \iff F \text{ is a polynomial automorphism.}$$

Thus, among Keller maps, strict containment $I_R(F) \subsetneq I_\Delta$, equivalently $\mathrm{Obs}(F) \neq 0$, is precisely the counterexample condition.

We organize this strict containment by generic degree. On coordinate rings, F induces

$$F^* \colon \mathbb{C}[t_1, \ldots, t_n] \longrightarrow A, \qquad t_i \longmapsto F_i.$$

Let

$$B = \mathrm{im}(F^*) = \mathbb{C}[F_1, \ldots, F_n] \subseteq A, \qquad K = \mathrm{Frac}(B), \qquad L = \mathrm{Frac}(A).$$

For a Keller map, the F_i are algebraically independent, so F^* identifies the target coordinate ring with B. Thus $K = \mathbb{C}(F_1, \ldots, F_n) \subseteq L = \mathbb{C}(x_1, \ldots, x_n)$ is the induced finite separable extension of function fields, and $[L : K]$ is the generic degree. If L/K is normal, the classical Galois case of the Jacobian conjecture implies that F is a polynomial automorphism. Since extensions of degree one and separable extensions of degree two are normal, neither degree can occur for a counterexample. Generic degree three is therefore the first possible counterexample degree.

We henceforth study polynomial maps $F \colon \mathbb{A}^3_{\mathbb{C}} \to \mathbb{A}^3_{\mathbb{C}}$ with generic degree $[L : K] = 3$. Here $JC(3)$ denotes the conjecture in three variables, while $[L : K] = 3$ is the separate generic-degree condition. Since polynomial automorphisms have generic degree one, every map with $[L : K] = 3$ is already nonautomorphic. In this generic-degree-three case,

$$F \text{ is a counterexample to } JC(3) \quad \iff \quad F \text{ is Keller.}$$

Once $[L : K] = 3$ is assumed, the residual decomposition is field-theoretic. The generic collision algebra has a diagonal factor L and a quadratic residual factor D_α:

$$K \otimes_B C_F \simeq_K L \times D_\alpha.$$

The quadratic residual factor is split in the normal branch and is a field in the nonnormal branch. In the latter case it is L-isomorphic to the normal closure N. The Keller condition identifies the counterexample locus and, through Galois rigidity, excludes the normal branch. Hence every generic-degree-three counterexample to $JC(3)$ has

$$K \otimes_B C_F \simeq_K L \times N, \qquad \mathrm{Gal}(N/K) \cong S_3.$$

Section 2 develops the dimension-independent collision obstruction, generic base change, marked-root decomposition, and automorphism criterion. Section 3 specializes these constructions to cubic extensions. The closing low-degree corollary records the input used by the planar sequel.

2 General construction

For a fixed dimension n, where can $JC(n)$ fail? We begin with an arbitrary polynomial map. Its collision ideal is canonically contained in the diagonal ideal, and $\mathrm{Obs}(F) = I_\Delta/I_R(F)$ measures their failure to agree. Once F is assumed Keller, strict containment is exactly the counterexample condition. No Keller hypothesis enters the collision construction.

2.1 The two ideals and their obstruction

Throughout the paper, the base field is $\mathbb{C}$. Let

$$F = (F_1, \dots, F_n) \colon X = \mathbb{A}^n_{\mathbb{C}} \longrightarrow \mathbb{A}^n_{\mathbb{C}}$$

be a polynomial map, and let

$$A = \mathbb{C}[x_1, \dots, x_n], \qquad S = \mathbb{C}[x_1, \dots, x_n, y_1, \dots, y_n].$$

Define

$$I_R(F) = (F_i(x) - F_i(y))_{i=1}^n, \qquad I_\Delta = (x_i - y_i)_{i=1}^n, \qquad C_F := S/I_R(F), \qquad R_F := \operatorname{Spec} C_F.$$

For a fixed map F, we abbreviate $I_R(F)$ to I_R.

Proposition 2.1 (Canonical containment)**.** *For every polynomial map F,*

$$I_R(F) \subseteq I_\Delta.$$

Proof. For $P \in \mathbb{C}[x_1, \dots, x_n]$, the difference $P(x) - P(y)$ belongs to $(x_1 - y_1, \dots, x_n - y_n)$. This follows by induction on P, or by telescoping each monomial one coordinate at a time. Apply the statement to each coordinate F_i. □

Let $q_F \colon S \twoheadrightarrow C_F$ be the quotient map. Diagonal substitution $S \twoheadrightarrow A$, given by $s(x, y) \mapsto s(x, x)$, has kernel I_Δ. Since $I_R \subseteq I_\Delta$, it descends uniquely to a surjection

$$\bar{\mu}_F \colon C_F \twoheadrightarrow A, \qquad [s]_{I_R} \longmapsto s(x, x).$$

We define the ideal of the diagonal in R_F by

$$\mathrm{Obs}(F) := q_F(I_\Delta) = I_\Delta/I_R \subseteq C_F.$$

Proposition 2.2 (The diagonal kernel)**.** *As ideals of C_F,*

$$\ker(\bar{\mu}_F) = \mathrm{Obs}(F) = I_\Delta/I_R.$$

Equivalently,

$$\ker(\bar{\mu}_F) = \mathrm{Obs}(F) = 0 \quad \Longleftrightarrow \quad I_R = I_\Delta.$$

Proof. The construction of $\bar{\mu}_F$ identifies its kernel with $I_\Delta/I_R = \mathrm{Obs}(F)$. Since $I_R \subseteq I_\Delta$,

$$I_\Delta/I_R = 0 \quad \Longleftrightarrow \quad I_R = I_\Delta.$$

□

2.2 The fiber-product interpretation

Let

$$A = \mathbb{C}[x_1, \dots, x_n], \qquad B = \mathbb{C}[F_1, \dots, F_n] \subseteq A.$$

Let $X = \operatorname{Spec} A$, let $Y = \operatorname{Spec} B$, and let

$$f_F \colon X \longrightarrow Y$$

be the morphism induced by $B \hookrightarrow A$. The homomorphisms $\iota_x, \iota_y \colon A \rightrightarrows C_F$, defined by $\iota_x(a) = [a(x)]$ and $\iota_y(a) = [a(y)]$, agree on B because $[F_i(x)] = [F_i(y)]$ in C_F. The equality $\iota_x|_B = \iota_y|_B$ defines the B-algebra structure on C_F.

Theorem 2.3 (Fiber-product interpretation)**.** *There is a canonical isomorphism*

$$R_F \cong X \times_Y X$$

under which $\operatorname{Spec}(\bar{\mu}_F) \colon X \to R_F$ *is the relative diagonal* $\Delta_{X/Y}$*. Dually, there is a canonical* B*-algebra isomorphism*

$$C_F \xrightarrow{\sim}_B A \otimes_B A$$

under which $\bar{\mu}_F$ *is tensor multiplication* $a \otimes a' \mapsto aa'$*.*

Proof. For every commutative $\mathbb{C}$-algebra T, maps from either C_F or $A \otimes_B A$ to T are pairs $f, g \colon A \rightrightarrows T$ agreeing on B. In the presentation of C_F, this agreement is imposed by the relations $F_i(x) = F_i(y)$, since the F_i generate B. The common universal property gives the algebra isomorphism, and applying Spec gives the scheme isomorphism. Tensor multiplication corresponds to $\bar{\mu}_F$, so its dual is the relative diagonal. □

On complex points,

$$R_F(\mathbb{C}) = \{(u, v) : F(u) = F(v)\}.$$

Inside $X \times X = \operatorname{Spec} S$, the ideals I_R and I_Δ define $X \times_Y X$ and its diagonal. Thus

$$I_R = I_\Delta \quad \Longleftrightarrow \quad X \times_Y X = \Delta_{X/Y}(X)$$

as closed subschemes. This is the geometric form of obstruction vanishing.

2.3 Generic fibers and marked-root decomposition

Let F be a polynomial map. Retain A, and set

$$B = \mathbb{C}[F_1, \dots, F_n] \subseteq A.$$

Let

$$K = \operatorname{Frac}(B), \qquad L = \operatorname{Frac}(A).$$

Proposition 2.4 (Keller generic geometry)**.** *If* F *is Keller, then its coordinate functions are algebraically independent, the morphism*

$$f_F \colon \operatorname{Spec} A \longrightarrow \operatorname{Spec} B$$

is étale and quasi-finite, and L/K *is a finite separable extension. Consequently, every Keller map, and in particular every complex Jacobian-conjecture counterexample, has a finite separable generic field extension.*

Proof. The identity

$$dF_1 \wedge \cdots \wedge dF_n = \det(JF)\, dx_1 \wedge \cdots \wedge dx_n \neq 0$$

gives algebraic independence, so $B \cong \mathbb{C}[u_1, \ldots, u_n]$. As a B-algebra, A has the presentation

$$A \cong \mathbb{C}[u_1, \ldots, u_n, X_1, \ldots, X_n]/(F_i(X) - u_i)_{i=1}^n.$$

Its relative Jacobian determinant is the unit $\det(JF)$. The Jacobian criterion therefore makes f_F étale, hence quasi-finite. Since $B \subseteq A$, the morphism is dominant, so quasi-finiteness gives $[L : K] < \infty$. Its generic point is étale, and therefore L/K is separable. □

For the remainder of this subsection, assume that F is dominant and generically finite. Let

$$\theta \colon K \otimes_B A \longrightarrow L, \qquad \lambda \otimes a \longmapsto \lambda a.$$

Let

$$\mu_L \colon L \otimes_K L \longrightarrow L, \qquad \ell \otimes \ell' \longmapsto \ell\ell',$$

denote tensor multiplication.

Theorem 2.5 (Generic collision bridge)**.** *The map θ is an isomorphism. Consequently, there is a K-algebra isomorphism*

$$\Phi_F \colon K \otimes_B C_F \xrightarrow{\sim}_K L \otimes_K L,$$

and the following square commutes:

$$\begin{array}{ccc} K \otimes_B C_F & \xrightarrow{1 \otimes \bar{\mu}_F} & K \otimes_B A \\ {\scriptstyle \Phi_F}\downarrow & & \downarrow{\scriptstyle \theta}\ \sim \\ L \otimes_K L & \xrightarrow{\mu_L} & L. \end{array}$$

Proof. Let $U = B \setminus \{0\}$. Since $K = U^{-1}B$, localization gives

$$K \otimes_B A \cong U^{-1}A \subseteq L,$$

and under this identification θ is the inclusion. Since L/K is finite, each x_i is algebraic over K; hence

$$U^{-1}A = K[x_1, \ldots, x_n]$$

is a finite-dimensional domain over K, and therefore a field. It contains A, so it contains $\mathrm{Frac}(A) = L$. Thus $U^{-1}A = L$, and θ is an isomorphism.

Using Theorem 2.3 and standard tensor base change, define Φ_F by the composite

$$K \otimes_B C_F \cong K \otimes_B (A \otimes_B A) \cong (K \otimes_B A) \otimes_K (K \otimes_B A) \xrightarrow{\theta \otimes \theta} L \otimes_K L.$$

Under this composite, both routes around the square send $\lambda \otimes (a \otimes a')$ to $\lambda a a'$. Hence the square commutes. □

Suppose now that L/K is separable and has a power basis $L = K(\alpha)$. Let $m_\alpha(T) \in K[T]$ be the minimal polynomial of α, and let $h_\alpha(T) \in L[T]$ be determined by

$$m_\alpha(T) = (T - \alpha)h_\alpha(T) \qquad \text{in } L[T].$$

Set

$$D_\alpha := L[T]/(h_\alpha).$$

Theorem 2.6 (Marked-root tensor decomposition)**.** *Give $L \otimes_K L$ its L-algebra structure through the left tensor factor. There is an L-algebra isomorphism*

$$\Phi_\alpha \colon L \otimes_K L \xrightarrow{\sim}_L L \times D_\alpha,$$

under which μ_L is the first-coordinate map.

Proof. View m_α in $L[T]$. Separability gives $h_\alpha(\alpha) = m'_\alpha(\alpha) \neq 0$, so $T - \alpha$ and h_α are coprime. The power-basis isomorphism and the Chinese remainder theorem give the composite

$$L \otimes_K L \xrightarrow{\sim}_L L[T]/(m_\alpha) \xrightarrow{\sim}_L L[T]/(T-\alpha) \times L[T]/(h_\alpha) \xrightarrow{\sim}_L L \times D_\alpha.$$

The first coordinate of the composite is evaluation at $T = \alpha$, which is precisely μ_L. □

Together with Theorem 2.5, this gives

$$K \otimes_B C_F \simeq_K L \times D_\alpha,$$

with the base-changed diagonal evaluation equal to the first-coordinate map.

2.4 Automorphism detection

Theorem 2.7 (Automorphism criterion)**.** *For an arbitrary polynomial map $F \colon \mathbb{A}^n_{\mathbb{C}} \to \mathbb{A}^n_{\mathbb{C}}$, the following are equivalent:*

(1) *F is a polynomial automorphism;*

(2) *F is injective on $\mathbb{A}^n(\mathbb{C})$;*

(3) *$I_R(F) = I_\Delta$;*

(4) *$\mathrm{Obs}(F) = 0$.*

Proof. Suppose first that F has polynomial inverse $G = (G_1, \ldots, G_n)$. Then

$$x_i - y_i = G_i(F(x)) - G_i(F(y)) \in I_R$$

for every i, so $I_\Delta \subseteq I_R$. Proposition 2.1 gives equality. Thus (1) implies (3).

Suppose $I_R = I_\Delta$, and let $\mathrm{ev}_{(a,b)} \colon S \to \mathbb{C}$ be evaluation at (a, b). If $F(a) = F(b)$, then

$$I_R \subseteq \ker(\mathrm{ev}_{(a,b)}).$$

Therefore $x_i - y_i \in I_\Delta = I_R$ gives $a_i - b_i = 0$ for every i. Thus $a = b$, proving (3)⇒(2). The Ax–Grothendieck theorem [Ax69] gives (2)⇒(1). Finally, Proposition 2.2 gives (3)⟺(4). □

Applied to Keller maps, Theorem 2.7 says that $JC(n)$ is equivalent to universal vanishing of $\mathrm{Obs}(F)$, or equivalently to $I_R(F) = I_\Delta$.

We will also use the classical Galois case of the conjecture. It is the step that rules out the normal cubic branch under the Keller hypothesis.

Theorem 2.8 (Classical Galois rigidity)**.** *Let*

$$F\colon \mathbb{A}^n_{\mathbb{C}} \longrightarrow \mathbb{A}^n_{\mathbb{C}}, \qquad \det JF \in \mathbb{C}^\times,$$

and put

$$K = \mathbb{C}(F_1, \dots, F_n), \qquad L = \mathbb{C}(x_1, \dots, x_n).$$

If L/K is normal, then F is a polynomial automorphism and $L = K$.

Proof. The Keller condition makes L/K finite and separable. Thus normality makes it Galois, and the conclusion is the classical Galois case of the Jacobian conjecture due to Campbell, with algebraic treatments by Razar and Wright [Cam73, Raz79, Wri81]. □

Lean correspondence. `CollisionIdeals/General/Collision/Diagonal.lean`, `CollisionIdeals/General/Collision/Ideal.lean`, `CollisionIdeals/General/Collision/CollisionDiagonal.lean`, `CollisionIdeals/General/FiberProduct/Polynomial.lean`, `CollisionIdeals/General/GenericFiber/FunctionField.lean`, `CollisionIdeals/General/GenericFiber/CollisionDiagonal.lean`, `CollisionIdeals/General/GenericFiber/MarkedRootDecomposition.lean`, `CollisionIdeals/General/Keller/Etale.lean`, `CollisionIdeals/General/Keller/GenericFiniteness.lean`, `CollisionIdeals/General/Automorphism/Equivalences.lean`, and `CollisionIdeals/General/Automorphism/GaloisRigidity.lean`. The polynomial fiber-product module uses the abstract target polynomial ring. The generic-fiber modules construct the equivalence over the image algebra $B = \mathbb{C}[F_1, \dots, F_n]$ and its generic base change, proving surjectivity of θ from positive finite rank and identifying diagonal evaluation with multiplication. The general Keller modules prove étaleness over the coordinate-image algebra and finite separability of the induced function-field extension directly from `IsKeller`; flatness follows from étaleness. Ax–Grothendieck remains an explicit parameter of the Lean automorphism equivalence.

3 The cubic S_3 obstruction in dimension three

We now specialize the dimension-independent constructions of Section 2 to generic degree three. After the diagonal is exhibited as the first factor, the complementary generic collision algebra has degree two over L. For a map of generic degree three it is either split, corresponding to a normal cubic extension, or connected, corresponding to a nonnormal cubic extension. Degree three already rules out polynomial automorphy, so the counterexamples to $JC(3)$ in generic degree three are precisely the Keller maps. For a Keller map, however, the split branch is impossible: it makes L/K normal and hence, by Theorem 2.8, forces F to be a polynomial automorphism, contradicting $[L : K] = 3$. Hence only the connected branch remains; its residual factor is the normal closure, with Galois group S_3.

3.1 The cubic extension and its normal closure

Let

$$F = (F_1, F_2, F_3)\colon \mathbb{A}^3_{\mathbb{C}} \longrightarrow \mathbb{A}^3_{\mathbb{C}}$$

be dominant and generically finite. Let

$$A = \mathbb{C}[x_1, x_2, x_3], \qquad B = \mathbb{C}[F_1, F_2, F_3] \subseteq A, \qquad K = \mathrm{Frac}(B), \qquad L = \mathrm{Frac}(A).$$

Assume that F has generic degree three:

$$[L : K] = 3.$$

Because K has characteristic zero, L/K is separable. Choose a primitive element α [The26e], so that $L = K(\alpha)$ with power basis $1, \alpha, \alpha^2$. Retain the notation of Theorem 2.6: if $m_\alpha \in K[T]$ is the minimal polynomial of α, then

$$m_\alpha(T) = (T - \alpha)h_\alpha(T) \qquad \text{in } L[T].$$

The residual factor in the marked-root decomposition is

$$D_\alpha = L[T]/(h_\alpha).$$

Since $[L : K] = 3$, it has dimension two over L. Let N/K be the normal closure of L/K, containing L through the marked inclusion. Since L/K is finite separable, N/K is Galois [The26c]. Let

$$G = \mathrm{Gal}(N/K), \qquad H = \mathrm{Gal}(N/L) = \{g \in G : g|_L = \mathrm{id}_L\},$$

where H fixes the marked copy of L in N. Restriction of automorphisms gives a bijection

$$G/H \xrightarrow{\sim} \mathrm{Hom}_K(L, N), \qquad gH \longmapsto g|_L.$$

Because the conjugates of L generate its normal closure N, an element of G fixing every conjugate embedding fixes N. Hence the coset action is faithful and

$$\mathrm{core}_G(H) = \bigcap_{g \in G} gHg^{-1} = 1.$$

The residual collision algebra detects which cubic branch occurs. We include the short argument in order to fix its marked identification with the normal closure.

Proposition 3.1 (Connected cubic residual factor)**.** *The following conditions are equivalent:*

(i) $\mathrm{Spec}\, D_\alpha$ *is connected;*

(ii) D_α *is a field;*

(iii) h_α *is irreducible over* L*;*

(iv) L/K *is nonnormal.*

When they hold, there is an L*-algebra isomorphism*

$$\varphi_\alpha \colon D_\alpha \xrightarrow{\sim}_L N.$$

Consequently,

$$\Psi_\alpha \colon L \otimes_K L \xrightarrow{\sim}_L L \times N,$$

under which μ_L *is the first-coordinate map.*

Proof. Write

$$m_\alpha(T) = T^3 + a_2T^2 + a_1T + a_0.$$

The quadratic h_α is separable, so D_α is finite étale over L; finite étale algebras over a field are finite products of finite separable field extensions [The26a]. If h_α is irreducible, then D_α is a field and hence has connected spectrum. If it is reducible, it is a product of two distinct linear factors; the Chinese remainder theorem then gives $D_\alpha \simeq_L L \times L$, whose spectrum is disconnected. This proves the equivalence of (i), (ii), and (iii).

If h_α is reducible, it has a root $\beta \in L$. Separability gives $\beta \neq \alpha$, and the third root is

$$\gamma = -a_2 - \alpha - \beta \in L.$$

Thus m_α would split over L. Since $L = K(\alpha)$, the standard generator criterion for normality [The26d] shows that L/K would be normal. Conversely, if L/K is normal, then m_α splits over L, so h_α is reducible. Hence (iii) and (iv) are equivalent.

Assume these equivalent conditions, and let β denote the image of T in D_α. This field contains the three roots

$$\alpha, \qquad \beta, \qquad -a_2 - \alpha - \beta$$

of m_α, and is generated over L by β. It is therefore the splitting field of m_α, which is the normal closure N of L/K [The26f]. Choose the resulting L-algebra isomorphism

$$\varphi_\alpha \colon D_\alpha \xrightarrow{\sim}_L N.$$

Let

$$\Phi_\alpha \colon L \otimes_K L \xrightarrow{\sim}_L L \times D_\alpha$$

be the marked-root decomposition of Theorem 2.6. Under this isomorphism, μ_L is the first-coordinate map. Composing its second factor with φ_α, let

$$\Psi_\alpha = (\mathrm{id}_L \times \varphi_\alpha) \circ \Phi_\alpha.$$

Since $\mathrm{id}_L \times \varphi_\alpha$ leaves the first factor unchanged, the same description of μ_L holds under Ψ_α. □

Remark (Ordered roots). Let

$$m_\alpha(T) = T^3 + a_2T^2 + a_1T + a_0$$

have roots $\alpha_1, \alpha_2, \alpha_3$ in N. For distinct i, j, k,

$$\alpha_k = -a_2 - \alpha_i - \alpha_j, \qquad K(\alpha_i, \alpha_j) = K(\alpha_1, \alpha_2, \alpha_3) = N.$$

In the nonnormal branch, the residual factor marks two distinct roots and is therefore already the ordered-root normal closure.

Lean correspondence. `CollisionIdeals/ComplexThree/Cubic/ResidualCriteria.lean` proves the four-way equivalence of Proposition 3.1 as `cubicResidual_tfae`, independently of Keller or rigidity. `CollisionIdeals/ComplexThree/Cubic/Canonical.lean` packages the Keller specialization of Theorem 3.3 as `complexThreeGenericDegreeThreeS3Collision`. From Keller, generic degree three, and the explicit input `ComplexKellerGaloisRigidity 3`, it constructs the marked normal closure and cubic data, and proves diagonal compatibility, obstruction nonvanishing, and the S_3 identification for the same residual field. `CollisionIdeals/ComplexThree/Cubic/Criteria.lean` proves the Keller/counterexample equivalence in generic degree three. Finiteness of a supplied normal closure is derived in `CollisionIdeals/General/Galois/FiniteNormalClosure.lean`, adapted from the reviewed `StacksCore` proof of tag 09DT [Axi26]. Object-level parity is recorded in `SEMANTIC-PARITY.md`.

Corollary 3.2 (The generic-degree-three counterexample locus)**.** *The map F is not a polynomial automorphism. Equivalently, on this generic-degree-three locus,*

$$F \text{ is a counterexample to } JC(3) \quad \Longleftrightarrow \quad F \text{ is Keller.}$$

When these equivalent conditions hold, L/K is nonnormal and $\operatorname{Spec} D_\alpha$ *is connected.*

Proof. If F were a polynomial automorphism, its coordinate functions would generate A, so $B = A$ and $L = K$, contrary to $[L : K] = 3$. The displayed equivalence now follows from the definition of a counterexample to $JC(3)$.

Assume these conditions, so that F is Keller. If L/K were normal, Theorem 2.8 would give $L = K$, again contradicting $[L : K] = 3$. Proposition 3.1 now identifies nonnormality with connectedness of $\operatorname{Spec} D_\alpha$. □

Theorem 3.3 (Generic cubic S_3 collision)**.** *Assume that* $\operatorname{Spec} D_\alpha$ *is connected. There is a K-algebra isomorphism*

$$\Psi_F \colon K \otimes_B C_F \xrightarrow{\sim}_K L \times N$$

under which, after the identification $K \otimes_B A \simeq_K L$ of Theorem 2.5, the base-changed diagonal evaluation

$$1 \otimes \bar{\mu}_F \colon K \otimes_B C_F \longrightarrow K \otimes_B A$$

is the first-coordinate map. In particular,

$$\Psi_F(\ker(1 \otimes \bar{\mu}_F)) = 0 \times N, \qquad \operatorname{Gal}(N/K) \cong S_3,$$

and

$$\operatorname{Obs}(F) \neq 0, \qquad I_R \subsetneq I_\Delta.$$

By Corollary 3.2, the connectedness hypothesis holds whenever F is Keller; in generic degree three these Keller maps are precisely the counterexamples to $JC(3)$.

Proof. Proposition 3.1 gives an L-algebra isomorphism

$$\Psi_\alpha \colon L \otimes_K L \xrightarrow{\sim}_L L \times N$$

under which μ_L is the first-coordinate map. Theorem 2.5 gives a K-algebra isomorphism

$$\Phi_F \colon K \otimes_B C_F \xrightarrow{\sim}_K L \otimes_K L$$

which carries the base-changed diagonal evaluation to μ_L. After restricting Ψ_α from L to K, let

$$\Psi_F = \Psi_\alpha \circ \Phi_F.$$

The asserted description of diagonal evaluation follows, and hence

$$\Psi_F(\ker(1 \otimes \bar{\mu}_F)) = 0 \times N \neq 0.$$

If $\bar{\mu}_F \colon C_F \to A$ were injective, flat base change along the localization $B \to K$ would make $1 \otimes \bar{\mu}_F$ injective. This would contradict the displayed kernel. Thus $\ker(\bar{\mu}_F) \neq 0$, so Proposition 2.2 gives $\operatorname{Obs}(F) \neq 0$, and Proposition 2.1 gives $I_R \subsetneq I_\Delta$.

For the Galois group, Proposition 3.1 identifies N with the degree-two field D_α over L. The fixed-field degree formula [The26b] gives

$$|H| = [N : L] = 2, \qquad [G : H] = [L : K] = 3.$$

The normal-closure identities recalled above give $\mathrm{core}_G(H) = 1$. Thus the action on the three cosets of H embeds G into S_3. Since $|G| = [G : H]|H| = 3 \cdot 2 = 6$, the faithful action is an isomorphism:

$$\mathrm{Gal}(N/K) \cong S_3.$$

□

4 Conclusion

The collision construction encodes injectivity by the ideal $\mathrm{Obs}(F) \subseteq C_F$. This ideal is the kernel of diagonal evaluation, and its vanishing is equivalent to the scheme-theoretic equality $R_F = \Delta$. After passage to the generic fiber, that diagonal becomes the multiplication factor of $L \otimes_K L$, while the remaining factors record the other embeddings of L.

In generic degree three, the diagonal persists as the first factor beside a quadratic residual algebra, and nonautomorphy is automatic. Thus the counterexample condition is equivalent to the Keller condition. On this locus the classical Galois case rules out the split, normal branch; hence the residual algebra is connected, it is the normal closure N, its Galois group is S_3, and

$$K \otimes_B C_F \simeq_K L \times N.$$

Counterexamples of other generic degrees lie outside the present analysis.

Relation to explicit constructions. Alpöge's example has generic degree three [Gao26, Theorem 3.3], so Theorem 3.3 applies to it. Van den Essen [vdE26, Sections 2–5] obtains a linearly equivalent map by prescribing one component and seeking the other two linear in one source variable.

Tao's exposition realizes this example through multiplication of a linear and a quadratic binary form [Tao26]. Generic collisions are pairs of such factorizations of the same cubic. Repeating the marked linear factor gives the diagonal factor L; choosing either remaining root gives D_α. This is the marked-root decomposition underlying our generic collision algebra, with $D_\alpha \simeq_L N$ in the nonnormal branch.

We distinguish three objects: an individual map F of generic degree three; a collection of candidate maps specified by a coordinate ansatz; and the normal closure attached to each Keller member of generic degree three. Our theorem applies independently of the ansatz. For each such member, the marked field tower satisfies

$$L = N^H, \qquad [L : K] = 3, \qquad [N : L] = 2, \qquad [N : K] = 6.$$

Here $G = \mathrm{Gal}(N/K) \cong S_3$, and $H = \mathrm{Gal}(N/L) \cong S_2$ stabilizes the marked embedding. The residual factor is the degree-six field N, not a parameter space of maps. The three cosets G/H index the K-embeddings of this fixed L into N: they are the geometric generic sheets of one map, not members of the ansatz-defined collection.

The formalization uses Lean 4.28.0 and Mathlib v4.28.0 [dMU21, The20]. Its Palomar-facing declaration is `CollisionIdeals.Palomar.PaperOne.genericDegreeThreeS3Collision`, checked by `lake build CollisionIdeals.Palomar.PaperOne.Solution`.

Corollary 4.1 (First possible generic degree)**.** *If F is Keller and $[L : K] \leq 2$, then F is a polynomial automorphism. Consequently, every complex Jacobian-conjecture counterexample has generic degree at least three.*

Proof. Proposition 2.4 makes L/K finite and separable. In degree one it is trivial, and in degree two it is Galois. Theorem 2.8 applies in either case. □

Paper II specializes this low-degree boundary to ambient dimension two and makes the quadratic collision contradiction explicit through the planar secant projector.

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